

ORIGINAL ARTICLE

Meta-Analysis of Genome-Wide Association Studies of Doxorubicin-Induced Arrhythmia Identifies *MMP16* and *KCTD15* as Risk Loci

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BACKGROUND: Anthracyclines are a widely used and effective class of chemotherapy. However, a major limitation for their use is cardiotoxicity, manifesting as systolic dysfunction, congestive heart failure, and arrhythmias. The objective of this study was to identify genes and genetic risk variants for doxorubicin-induced arrhythmia.

METHODS: We performed genome-wide association studies and meta-analysis across 2 independent nonoverlapping, genetically homogenous data sets from the Molecular Epidemiology Resource (cases N=77, controls N=1184) and Tapestry (cases N=13, controls N=172). Doxorubicin-related arrhythmia was the primary cardiac outcome, followed by investigation of a broader phenotype of cardiac events. Effects of doxorubicin on top-associated genes were assessed by quantitative polymerase chain reaction on RNA extracted from human cardiomyocytes following treatment with doxorubicin and by echocardiography in a zebrafish model of doxorubicin-induced cardiomyopathy.

RESULTS: In the meta-analysis, rs12151014 at the *KCTD15* locus was associated with doxorubicin-related arrhythmia at the genome-wide significance (minor allele frequency in Molecular Epidemiology Resource cases 0.195 versus 0.069 in controls; Tapestry cases 0.192 versus 0.079 in controls) $P=9.01 \times 10^{-9}$; odds ratio, 3.61. *MMP16* rs1094199 was associated with the broader phenotype of arrhythmia and heart failure (minor allele frequency in Molecular Epidemiology Resource cases 0.079 versus 0.019 in controls; Tapestry cases 0.038 versus 0.015) $P=1.98 \times 10^{-8}$; odds ratio=6.15. In human cardiomyocytes, *MMP16* and *KCTD15* gene expression was significantly downregulated by doxorubicin, and in a zebrafish *MMP16* knockout was protective of doxorubicin-induced decline in left ventricular ejection fraction.

CONCLUSIONS: Genetic variants at *KCTD15* and *MMP16* are associated with doxorubicin-related arrhythmia and cardiac events. *MMP16* is a potential cardioprotective therapeutic target.

Key Words: anthracycline ■ atrial fibrillation ■ cardiotoxicity ■ heart failure ■ smoking

Anthracyclines are a widely used class of chemotherapy that are associated with cytotoxicity through double-strand DNA breaks. This class of chemotherapy is highly effective and is used for the treatment of several malignancies, including lymphoma, leukemia, Wilms' tumor, and breast cancer, small cell lung, endometrial, gastric, liver, kidney, ovarian, thyroid, and pediatric

cancers.¹ Exposure to anthracyclines is most common in patients with breast cancer and lymphoma. The major limitation for use is cardiotoxicity,^{2–5} manifesting as systolic dysfunction, congestive heart failure and arrhythmias, including atrial fibrillation (AF), ventricular tachycardia (VT), and heart block.^{6–8} The incidence and timing of arrhythmias in patients with or without cardiomyopathy

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCGEN.125.005651>.

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Circulation: Genomic and Precision Medicine is available at www.ahajournals.org/journal/circgen

Nonstandard Abbreviations and Acronyms	
AF	atrial fibrillation
AIC	anthracycline-induced cardiotoxicity
BMI	body mass index
CVD	cardiovascular disease
EF	ejection fraction
GWAS	genome-wide association study
KCTD15	potassium channel tetramerization domain containing 15
LD	linkage disequilibrium
MER	Molecular Epidemiology Resource
OR	odds ratio
sgRNA	single guide RNA
sgRNP	sgRNA–Cas9 ribonucleoprotein
TPM	transcripts per million
VT	ventricular tachycardia

are not well known.⁸ Although anthracyclines are known to be pro-arrhythmogenic,^{7,8} safety analyses in clinical trials have been primarily focused on outcomes of congestive heart failure and left ventricular systolic dysfunction specifically, often overlooking arrhythmia-specific end points.^{9–13} Nonetheless, reporting of arrhythmia is important, as AF is associated with stroke risk and generally requires anticoagulation, and ventricular arrhythmias and advanced heart block may be life-threatening.

The MER (Molecular Epidemiology Resource) prospective observational cohort study of newly diagnosed lymphoma characterized the cumulative incidence and risk factors of cardiovascular disease (CVD) in patients with lymphoma.² The MER study reported increasing age, male sex, current smoking, body mass index (BMI) and any anthracycline treatment to be significantly associated with risk of CVD and anthracyclines were associated with increased risk of congestive heart failure (2.71; $P<0.001$) and arrhythmia (heart failure=1.61; $P<0.01$), but not with valvular heart disease or acute coronary syndrome after adjustment for CVD risk factors.

A number of other, nonanthracycline-related cardiomyopathies, channelopathies and arrhythmias are known to have a genetic component,^{14–16} and the architecture is complex.^{17–19} As such, we hypothesized that genome-wide association study (GWAS) analysis could identify genetic variants for anthracycline-induced arrhythmia. Defining such variants has clinical importance, allowing for a personalized medicine approach to identify high-risk individuals for cardiac complications, who may then be better targeted for preventative measures, cardiac optimization, and close surveillance. Therefore, in the present study, we performed GWAS and meta-analysis across 2 independent cohorts, specifically for anthracycline-related cardiac events, including arrhythmia.

METHODS

Human studies were approved by the Mayo Clinic institutional review board, and subjects gave informed consent.

Animal study procedures were followed in accordance with Mayo Clinic Institutional guidelines.

Data Availability

Project Generation data were used under license, and thus are not publicly available due to use restrictions. Reasonable requests for Project Generation data from qualified researchers may be directed to Dr Jim Cerhan, Cerhan.James@Mayo.edu. Access to de-identified data will require a legal agreement and the permission of Regeneron.

Tapestry data were used under license, and thus are not publicly available due to use restrictions. Reasonable requests for Tapestry data from qualified researchers may be directed to Dr Konstantinos Lazaridis, lazaridis.konstantinos@mayo.edu. Access to de-identified data will require a legal agreement and the permission of Helix Inc.

Patient Populations

Clinical and genetic data were obtained through the MER² and Tapestry^{20,21} studies. In both data sets, participants with CVD (including arrhythmias) before lymphoma or breast cancer diagnoses were excluded from analyses.

The MER study characterized the cumulative incidence and risk factors for CVD, consisting of congestive heart failure, acute coronary syndrome, valvular heart disease and arrhythmia, in 3063 newly diagnosed lymphoma patients (>18 years) who were enrolled at the Mayo Clinic between 2002 and 2015. Genetic data were available for 2592 MER participants (Supplemental Methods; Figure S1). Genetic data for MER participants were generated through Project Generation, a collaboration between Mayo Clinic (Center for Individualized Medicine and Mayo investigators) and the Regeneron Genetics Center, which sequenced over 100 000 Mayo Clinic participants under institutional review board approval. This included whole exome sequencing, genotyping-by-sequencing SNP data, and subsequent genotype imputation. The sequencing platform was based on a modified TWIST Comprehensive Exome capture design that included additional backbone regions to capture common tagging variants across the genome.

These backbone variants, together with exome data, were used for genotype imputation to the TOPMed reference panel, enabling genome-wide coverage of both common and low-frequency variants. Exome sequencing data were used for sample-level quality control, while genome-wide imputed genotype data were used for all downstream association analyses.

Tapestry^{20,21} is a large-scale, decentralized and clinical-grade genomic sequencing study that recruited patients receiving care at Mayo Clinic. Tapestry enrolled >98 000 participants between July 2020 and May 2024. Exome+® sequencing^{22,23} was conducted at Helix Inc (San Mateo, CA), with CDC Tier 1 genomic results returned to participants and raw data stored at Mayo Clinic for future research use. The goal of Tapestry was to advance clinical practice, research discovery and genomic education. Written informed consent was obtained for all participants and approved by the Mayo Clinic IRB. In the current study, Tapestry patients with diagnoses of lymphoma or breast

cancer and available Exome+® data who received anthracycline therapy were identified (Figure S1). These patients were then curated by a team of cardiologists and internal medicine physicians (M.P., M.A., R.A., and C.A.) for demographic data (age at anthracycline dosing, sex, BMI, and diabetes) and cardiac events of heart failure or arrhythmia (AF, VT and complete heart block) that were attributed to anthracycline treatment. AF was defined based on presence of irregular R-R intervals with absence of distinct P waves and presence of fibrillatory waves on standard 12-lead ECG²⁴ sustained VT was defined as run of consecutive ventricular beats (monomorphic or polymorphic) for greater than (for sustained) 30 seconds on telemetry or ECG, or termed nonsustained VT if <30 seconds of duration.²⁵ Complete heart block was defined as no evidence of atrioventricular contraction detected by ECG (3) at any time within follow-up after the first anthracycline dose.²⁶

Quality Control of Genetic Data

Genotypes from exome sequencing, as well as those from imputation, were obtained from MER and Tapestry studies. Samples were evaluated for call rate, sex check, relatedness and population substructure using biallelic variants having a genotyping rate $\geq 98\%$, minor allele frequency $\geq 2\%$, and Hardy-Weinberg equilibrium P value $> 5 \times 10^{-8}$ while excluding those in highly volatile regions of the genome using Plink.²⁷ Samples with a call rate $\geq 98\%$ and inbreeding coefficients on the X-chromosome ≤ 0.3 for females and 0.7 or greater for males were retained. Relatedness was evaluated using KING-robust²⁸ implemented in Plink v2.²⁹ Among samples that were related up to the third degree (kinship coefficient ≥ 0.0442), only 1 sample from each pair or family was retained. Population substructure was evaluated using Eigenstrat.³⁰ Eigenstrat was set to remove population outliers 6 standard deviations from the mean over 5 iterations.

Following QC procedures within each cohort, genotype data from both studies were combined to evaluate and identify relateds between MER and Tapestry. Using a common set of variants, relatedness was evaluated using KING-robust in Plink v2 and related samples excluded, as described above.

Exome sequencing data were used for sample-level quality control, after which genome-wide imputed genotype data were used for all downstream association analyses. Functional annotation of variants was performed using ANNOVAR,³¹ and potential deleteriousness was assessed using Combined Annotation Dependent Depletion scores.

Following quality control, 2478 participants from the MER cohort and 1049 participants from the Tapestry cohort were retained for downstream analyses. Exclusions included samples with sex mismatch, excess relatedness, or population outlier status as defined above. Additionally, 16 related individuals across cohorts (12 duplicates and 4 first-degree relatives) were identified and excluded.

Imputation

Genotype imputation was not performed as part of the present study. Instead, preimputed genotype data were obtained for both the MER and Tapestry cohorts. For the MER cohort, imputation was performed by the Regeneron Genetics Center using Eagle (v2.4) for phasing and Minimac4 (v4.1.6) for imputation to the TOPMed reference panel (release R3). Imputed

genotype data for the Tapestry cohort were generated using a comparable pipeline.²⁰ Genotypes from imputation for variants with an $R^2 \geq 0.9$ and a minor allele frequency ≥ 0.001 were imported into Plink v2 for GWAS.

Genome-Wide Association Study and Meta-Analysis

Genome-wide association analyses were conducted separately within each cohort for 2 phenotypes: (1) arrhythmia and (2) a composite cardiac event defined as arrhythmia and heart failure following anthracycline exposure. Multivariable logistic regression was performed under an additive genetic model in PLINK v2,²⁹ with SNP dosage derived from imputed genotype data as the predictor. Although genetic data were generated using an exome-based sequencing platform, inclusion of backbone variants and subsequent imputation to the TOPMed reference panel enabled genome-wide coverage of both common and low-frequency variants. Association analyses were therefore conducted on genome-wide variation and were not restricted to coding regions.

Analyses were restricted to biallelic variants with imputation quality $R^2 \geq 0.9$ and minor allele frequency ≥ 0.001 . In the MER cohort, models were adjusted for age at cancer diagnosis, biological sex, radiation therapy (yes/no), and the first 3 principal components of ancestry (PC1–PC3). In the Tapestry cohort, models were adjusted for age, sex, and PC1–PC3; radiation therapy was not included as a covariate due to the limited sample size and number of events, which could lead to model instability.

Association results were generated separately for each phenotype within each cohort and combined using inverse-variance weighted, fixed effects meta-analysis implemented in PLINK v1.9. Estimates were used for primary inference to account for potential heterogeneity between cohorts. Between-study heterogeneity was assessed using the Cochran Q statistic and the I^2 metric.

Statistical significance was defined using the conventional genome-wide threshold ($P < 5 \times 10^{-8}$). Variants with $P < 1 \times 10^{-6}$ were considered suggestive and are reported in [Supplemental Material](#). Given the modest number of events, findings not meeting genome-wide significance were considered exploratory.

To identify independent association signals, linkage disequilibrium (LD)-based clumping was performed using PLINK v1.9. Variants were clumped based on meta-analysis P values, using imputed genotype data from the MER cohort as the LD reference panel. Clumping was performed with a 50 kb window ($-\text{clump-kb } 50$) and an LD threshold of $r^2 = 0.2$ ($-\text{clump-r2 } 0.2$). Within each locus, the variant with the lowest P value was designated as the index variant, and variants in LD above the specified threshold were assigned to that locus. Index variants are reported in Table 2 and [Table S1](#).

In Silico Analysis

The transcription factor binding affinity and the fold difference in binding between the 2 DNA sequences for the *KCTD15* promoter variants, rs62101381 and rs141116709, were predicted in silico using the Transcription Factor Affinity Prediction web tools,³² sTRAP difference between 2 sequences with matrix file transfac12.1, background model chordate conserved elements, and multiple test correction Benjamini-Hochberg.

Cardiomyocyte Culture

AC16 human cardiomyocytes were purchased from the American Tissue Culture Center (United States). Cardiomyocytes were maintained in DMEM/F-12 (DMEM/Nutrient Mixture F-12) medium (Thermo Fisher catalog no. 11320033) supplemented with 12.5% Fetal Bovine Serum and 1% Penicillin-streptomycin. Cells were incubated at 37°C in a humidified incubator containing 5% CO₂. The media was changed every 2 days, and cells were maintained for not >10 passages.

Quantitative Polymerase Chain Reaction Gene Expression

AC16 human cardiomyocytes were seeded at a density of 3×10³ cells per well in 6-well plates. After overnight incubation, the cells were treated with a vehicle control or 0.5 μmol/L Doxorubicin for 48 hours (6 replicates per group). Total RNA was extracted from the cardiomyocytes using the RNeasy Mini Kit (Qiagen, United States) following the manufacturer's instructions. RNA concentration and quality were determined using the NanoDrop 2000C spectrophotometer (Thermo Fisher Scientific, United States) and the Agilent Bioanalyzer 2100 (Agilent Technologies, United States), respectively. High-Capacity cDNA Reverse Transcription Kit (Thermo Fisher, catalog no. 4368813) was used to reverse transcribe each RNA sample into cDNA in a total volume of 20 μL. Using the cDNA samples, gene expression was assessed by quantitative polymerase chain reaction using Taqman Gene Expression Assay probes for MMP16 (Hs00234676_m1), KCTD15 (potassium channel tetramerization domain containing 15; Hs00225337_m1), and GAPDH (Hs99999905_m1), performed on a QuantStudio 3 Real-Time Polymerase Chain Reaction System (Applied Biosystems, United States). Gene expression levels of MMP16 and KCTD15 were assessed relative to GAPDH. Fold change was calculated using 2^{−ΔΔCT} formula.

In Vivo Model of Doxorubicin-Induced Cardiotoxicity

Animals

Zebrafish (*Danio rerio*) WIK strain was maintained under 14 hours light–10-hour dark cycle at 28.5°C and handled with care. The animal study protocols were approved by the Mayo Clinic Institutional Animal Care and Use Committee under protocol number A00002783-17.

Anthracycline-Induced Cardiotoxicity Model in Adult Zebrafish

To establish a progressive anthracycline-induced cardiotoxicity (AIC) model, we previously identified that a dose of 20 μg/g doxorubicin represents the highest concentration that did not induce significant fish mortality while still leading to reduced cardiac function. This dose aligns closely with those commonly used in rodent AIC models (15–25 mg/kg)³³ and approximates the cumulative dose threshold for cardiotoxicity in humans (550 mg/m², which is approximately equivalent to 15 mg/kg).³⁴ In this article, doxorubicin was administered via a single bolus of intraperitoneal injection, following the protocol previously described in our earlier publication.³⁵

Microhomology-Mediated End Joining-Based sgRNA Design and F₀ Genetic Testing

Single guide RNA (sgRNA) design and F₀ embryo injections were performed following a recently published microhomology-mediated end joining-based protocol.³⁶ In brief, target exon sequences were submitted to the MENTHU web-based platform (<http://genesculpt.org/menthu/>) to identify microhomology-mediated end joining target sites. sgRNA sequences with high MENTHU scores were selected for synthesis. Chemically modified sgRNAs were synthesized by Synthego (Synthego Corporation) and resuspended in nuclease-free duplex buffer (Integrated DNA Technologies, 11-01-03-01), then diluted to a final working concentration of 5 μmol/L. To generate sgRNP (sgRNA–Cas9 ribonucleoprotein) complexes, sgRNAs were incubated with Cas9 protein and microinjected into 1-cell stage zebrafish embryos. *mmp16a* and *mmp16b* were targeted independently in separate experiments to recapitulate the modifying effects of a genetic lesion in human patients in adult zebrafish. Injected F₀ embryos were individually collected for downstream analysis to assess knockout efficiency. Genomic DNA was extracted, and target regions were amplified and subjected to Sanger sequencing (Genewiz; <https://clims4.genewiz.com/CustomerHome/Index>). Resulting sequences were analyzed using the ICE v2 CRISPR Analysis Tool (Inference of CRISPR Edits; <https://www.synthego.com/products/bioinformatics/crispr-analysis>) to calculate knockout efficiency scores.³⁷ The primer sequences used to assess knockout efficiency were: *mmp16a*–MJ-F: GCATGTTTCAGATGCTCTGTCT R: AACTTCCTCAGCCCACAAGG *mmp16b*–MJ-F: TAGA GCCCATTTGTGTATTCTGR: ACTTGTTTAGATCCCTGTGATG.

In Vivo Echocardiography for Adult Fish Hearts

High-frequency echocardiography was performed at 28 days post doxorubicin injection using a Vevo 3100 imaging system (Fujifilm VisualSonics Inc., Toronto, Canada) equipped with a 50-MHz linear array transducer (MX700) following previously established methods.³⁸ Acoustic gel (Aquasonic 100, Parker Laboratories Inc) was applied to the transducer surface to ensure optimal coupling with the tissue interface. Zebrafish were anesthetized with tricaine (0.16 mg/mL) and positioned ventral side up on a soft-sponge stage. The MX700 transducer was aligned above the zebrafish to acquire sagittal imaging planes of the heart.³⁸ Three to 5 independent cardiac cycles per fish were measured to determine the average values. Cardiac function indices of the ejection fraction (EF) were calculated using the formula: EF=(EDV–ESV)/EDV. EDV and ESV are the ventricular volumes at the end-diastolic stage and end-systolic stage, respectively. Three to 5 consecutive cardiac cycles were analyzed for each fish, and the mean EDV, ESV, and derived EF were used for all subsequent analyses.

RESULTS

Patient Characteristics

Following QC and ancestry analysis, the MER lymphoma data set consisted of 1261 patients with lymphoma, 77 of whom had an outcome of arrhythmia and 1184 patients who did not experience arrhythmia after

Table 1. MER and Tapestry Participants Receiving Anthracycline, Number of Arrhythmia Cases and Controls, With Genotypic Data After QC (and Ancestry)

		MER data set			Tapestry data set		
		Arrhythmia	No arrhythmia	P value	Arrhythmia	No arrhythmia	P value
Baseline characteristics							
Cancer type	Breast cancer	0	0		7	118	
	Lymphoma	77	1184		6	54	
Age	Median (range)	65 (31–83)	56 (18–93)	2.00×10 ^{−88}	67 (43–89)	55 (18–79)	7.49×10 ^{−3}
Sex	Female (%)	21 (27)	506 (43)	0.029	7 (54)	145 (84)	1.02×10 ^{−2}
	Male (%)	56 (73)	678 (57)		6 (46)	27 (16)	
BMI	n	76	1160	0.46	13	171	0.11
	Median (range)	27.4 (19–45.4)	27.1 (16.9–49.6)		30 (22–64.2)	26.6 (18.8–51)	
Diabetes	No (%)	60 (78)	953 (80)		11 (85)	159 (92)	
	Yes (%)	6 (8)	72 (6)	0.53	2 (15)	13 (8)	0.33
	Missing (%)	11 (14)	159 (13)		0 (0)	0 (0)	
Radiation therapy	No (%)	67 (87)	991 (84)	0.88	12 (92)	166 (97)	0.46
	Yes (%)	10 (13)	193 (16)		1 (8)	6 (3)	
Outcomes							
Any arrhythmia	N	77	1184		13	172	
Atrial fibrillation	No				3	172	N/A
	Yes				10	...	
Ventricular tachycardia	No				7	...	N/A
	Yes				6	...	
Complete heart block	No				12	172	N/A
	Yes				1	...	
Heart failure	No (%)	55 (71)	1158 (98)	2.81×10 ^{−19}	12 (92)	167 (97)	0.37
	Yes (%)	22 (29)	26 (2)		1 (8)	5 (3)	

BMI indicates body mass index; and MER, Molecular Epidemiology Resource.

anthracycline treatment (Table 1). The mean follow-up time from diagnosis to arrhythmia event was 5.0 years, SD=3.6 years. The mean follow-up time from cancer diagnosis to the last follow-up was 9.3 years, SD=4.0 years for patients with an arrhythmia event and 7.2 years, SD=4.0 years for patients with no cardiovascular event. The Tapestry data set used for genetic analyses consisted of 125 patients with breast cancer and 54 patients with lymphoma, of which 13 patients had an outcome of arrhythmia and 172 patients who did not experience arrhythmia after anthracycline treatment (Table 1), yielding 90 unique arrhythmia cases (6.64%) and 1356 patients with no arrhythmia (93.4%) tested in the meta-analysis.

Type of arrhythmia was not collected in the MER study, but these data were available in Tapestry individuals, in which 10/13 patients experienced AF, 6/13 patients experienced VT and 1 patient experienced complete heart block attributed to doxorubicin. Four patients experienced more than 1 type of arrhythmia (at different time-points postdoxorubicin).

BMI measured, diabetes or radiation variables were not statistically different between patients with and without arrhythmia in either the MER or Tapestry

(Table 1) data sets. We observed significant differences for older age and male sex in both data sets, and these variables were incorporated into the final genetic model.

Not surprisingly, there was some overlap between arrhythmia and heart failure outcomes. In the larger MER data set, 22/77 (28.6%) patients who experienced arrhythmia also experienced heart failure, compared with 26/1158 (2.2%) of patients that did not experience arrhythmia, $P=2.81\times10^{-19}$. In the Tapestry data set, the same trend was observed, 1/13 (7.7%) versus 5/172 (2.9%), but did not reach statistical significance, $P=0.367$.

CHST8:KCTD15 Locus Reached Genome-Wide Significance for Association With Anthracycline-Related Arrhythmia

Meta-analysis of doxorubicin-related arrhythmia in the MER and Tapestry cohorts identified a locus on chromosome 19 reaching genome-wide significance ($P<5\times10^{-8}$; Figure 1A). Additional variants not meeting genome-wide significance are reported in the Supplemental Material (Table S1). The most significant association mapped

Table 2. Arrhythmia GWAS Summary Statistics From the MER and Tapestry Studies as Well as Meta-Analysis Results for Arrhythmia and Cardiac Event Phenotypes

Variant information					Minor allele frequencies of arrhythmia		Arrhythmia GWAS										Cardiac event GWAS	
							MER		Tapestry				Meta-analysis				Meta-analysis	
					Case	Con	Case	Con	OR	P value	OR	P value	OR	P value	OR	P value	OR	P value
CHST8;KCTD15	19	rs12151014	A	G	0.195	0.069	0.192	0.079	3.52	1.2×10 ⁻⁷	4.27	2.4×10 ⁻²	3.61	9.0×10⁻⁹	2.84	2.7×10 ⁻⁷		
PTPRD	9	rs62533683	C	T	0.065	0.012	0.077	0.012	7.66	1.1×10 ⁻⁶	17.91	2.3×10 ⁻²	8.32	9.2×10 ⁻⁸	6.18	1.2×10 ⁻⁶		
FOXB1;ANXA2	15	rs76263752	A	G	0.071	0.014	0.115	0.012	5.93	5.9×10 ⁻⁶	11.93	1.1×10 ⁻²	6.54	2.6×10 ⁻⁷	4.92	4.1×10 ⁻⁶		
MMP16	8	rs1094199	T	C	0.079	0.019	0.038	0.015	6.08	1.1×10 ⁻⁶	5.90	1.5×10 ⁻¹	6.07	3.7×10 ⁻⁷	6.15	2.0×10⁻⁸		
TMEM17;EHBP1	2	rs144003047	A	G	0.267	0.144	0.250	0.182	2.59	3.4×10 ⁻⁶	2.60	1.3×10 ⁻¹	2.59	1.0×10 ⁻⁶	1.99	9.1×10 ⁻⁵		
MMP16	8	rs60298754	T	C	0.078	0.020	0.038	0.017	5.39	4.2×10 ⁻⁶	4.56	2.1×10 ⁻¹	5.31	1.8×10 ⁻⁶	4.81	1.1×10 ⁻⁶		
MMP16	8	rs80023829	A	T	0.065	0.016	0.038	0.018	5.98	5.9×10 ⁻⁶	5.35	1.7×10 ⁻¹	5.91	2.2×10 ⁻⁶	5.04	3.7×10 ⁻⁶		
AKAP10	17	rs139899843	A	G	0.052	0.010	0.077	0.006	6.62	3.8×10 ⁻⁵	23.71	1.5×10 ⁻²	7.63	2.7×10 ⁻⁶	5.79	1.7×10 ⁻⁵		
PTPRN2	7	rs1628432	T	C	0.084	0.025	0.077	0.026	4.75	1.2×10 ⁻⁵	4.84	1.0×10 ⁻¹	4.76	3.1×10 ⁻⁶	3.20	2.7×10 ⁻⁴		
KITLG	12	rs116953322	G	A	0.097	0.025	0.038	0.015	4.44	4.8×10 ⁻⁶	3.22	3.6×10 ⁻¹	4.35	3.2×10 ⁻⁶	3.16	1.2×10 ⁻⁴		
LOC100288637	15	rs76267408	C	G	0.045	0.010	0.077	0.006	6.98	6.3×10 ⁻⁵	19.65	1.2×10 ⁻²	8.10	3.2×10 ⁻⁶	7.28	1.5×10 ⁻⁶		
LOC102724084	16	rs62049538	A	G	0.045	0.006	0.038	0.015	11.50	5.4×10 ⁻⁶	3.39	3.1×10 ⁻¹	9.38	4.9×10 ⁻⁶	7.13	1.6×10 ⁻⁵		

Independent variants ($r^2<0.2$) with a meta-analysis P value $<5\times10^{-6}$ in Arrhythmia GWAS are tabulated. Test of heterogeneity (Cochran Q statistic and I^2 heterogeneity index) showed no significant heterogeneity between the cohorts in either meta-analysis for all variants shown ($Q>0.35$, $P=0$). From left to right, gene or genes proximal to the variant, its chromosome, position, dbSNP ID (rsID), tested allele (A1), untested allele (A2), minor allele frequencies for arrhythmia cases and controls (Con) in MER and Tapestry, OR and P values (P) for arrhythmia GWAS from MER, Tapestry and meta-analysis results for the Arrhythmia GWAS and cardiac event are shown. Meta-analysis results (OR and P) were obtained using an inverse-variance weighted, fixed-effects model. Proximal genes and meta-analysis P for the top variant in both phenotypes are bolded. Cardiac event summary statistics from the MER and Tapestry studies are not shown. I^2 and Q values, along with summary statistics for all variants with meta-analysis $P<5\times10^{-5}$ for either phenotype, are shown in Table S1. GWAS indicates genome-wide association study;

MER, Molecular Epidemiology Resource; and OR, odds ratio.



to the 3' and 5' flanking regions of *CHST8* (carbohydrate sulfotransferase 8) and *KCTD15* (potassium channel tetramerization domain containing 15), respectively. Twelve variants at this locus were in complete LD ($r^2=1$), with P values ranging from 9.01×10^{-9} (sentinel variant, rs12151014) to 3.64×10^{-8} (Figure 1C; Table S2).

Functional annotation using ANNOVAR indicated that the sentinel variant and variants in strong LD at this locus are noncoding and map to the 5' flanking and promoter regions of *KCTD15*. No protein-altering variants were identified. Combined Annotation Dependent Depletion scores for these variants were consistent with potential regulatory effects. Although the associated region spans both *CHST8* and *KCTD15*, these data do not allow definitive assignment of a causal gene.

A detailed breakdown of variants at the *CHST8:KCTD15* locus is provided in Table S2. The sentinel variant, rs12151014, is located 6.1 kb upstream of *KCTD15* and 17.3 kb downstream of *CHST8*. In cohort-specific analyses, the minor allele frequency was 0.195 in cases versus 0.069 in controls in MER (odds ratio [OR]=3.52; $P=1.17\times10^{-7}$) and 0.19 versus 0.079 in Tapestry (OR=4.27; $P=0.024$). In meta-analysis, this variant demonstrated a genome-wide significant association ($P=9.01\times10^{-9}$; OR=3.61) with no evidence of heterogeneity (Cochran $Q=0.78$; $I^2=0$).

Other variants in complete LD with the sentinel variant mapped to the 5' flanking region of *KCTD15*, including rs62101381 and rs141116709, located 798 bp and 523 bp upstream of exon 1, respectively.

Suggestive Association at the *MMP16* Locus With Anthracycline-Related Arrhythmia

Variants at the *MMP16* (matrix metalloproteinase-16) locus on chromosome 8 demonstrated suggestive evidence of association ($P<1\times10^{-6}$; Table 2; Table S1) but did not meet the threshold for genome-wide significance. The lead variant, rs1094199, located in the 5' flanking region of *MMP16*, showed a meta-analysis association of $P=3.72\times10^{-7}$ (OR=6.07). (Figure 1A; Table S2). In cohort-specific analyses, the direction of effect was consistent across MER and Tapestry, with no evidence of heterogeneity (Cochran $Q=0.98$; $P=0$; Table S2). Additional loci with suggestive associations are reported in Table S1.

Overlap of Associated *CHST8:KCTD15* and *MMP16* Variants With Anthracycline-Related Heart Failure

Our available sample size for heart failure as an outcome related to anthracycline in the MER and Tapestry data sets was limited. However, given the relationship between arrhythmia and development of heart failure and vice versa,^{39,40} we also explored the *CHST8:KCTD15* and *MMP16* loci for association with a combined analysis of heart failure and arrhythmia, which we termed cardiac event. This analysis yielded a total number of 103 cases in the MER data set and 18 cases in the Tapestry data set. In this meta-analysis (Table 2, Table S3),

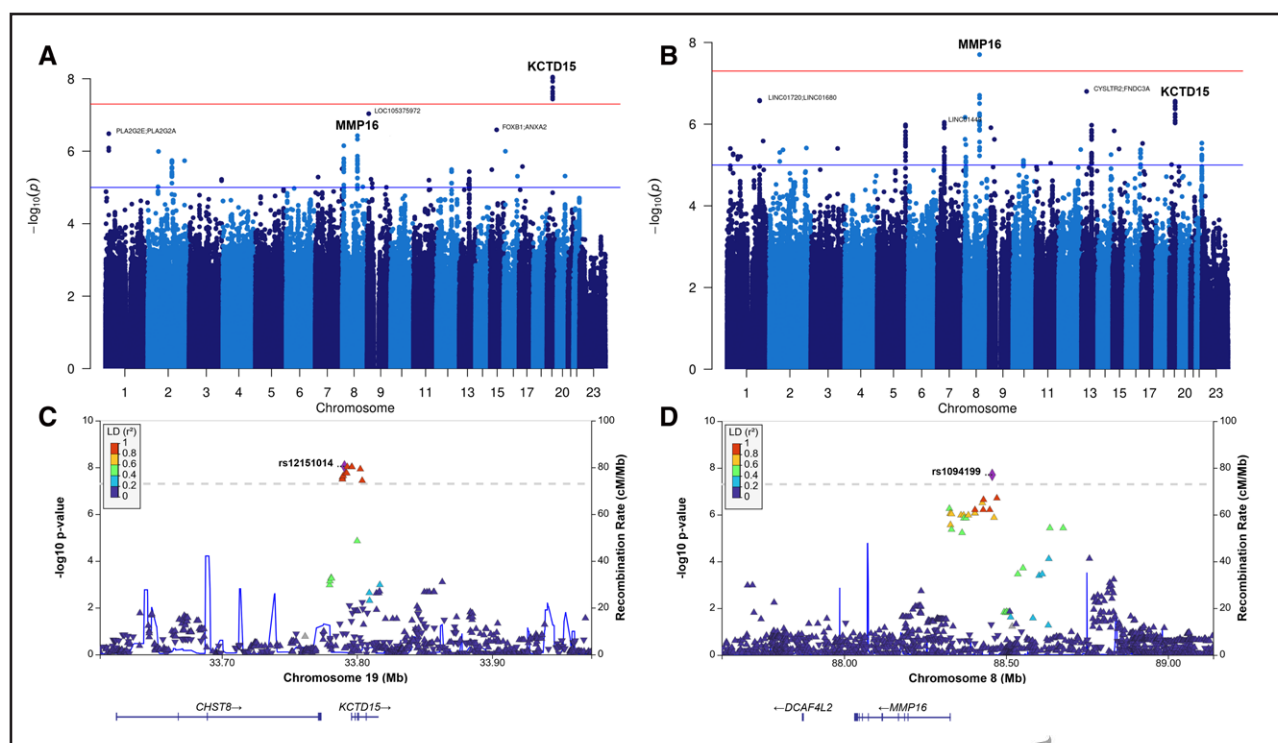


Figure 1. Meta-analysis of anthracycline-related genome-wide association study (GWAS) data sets.

A, Manhattan plot anthracycline-related arrhythmia. **B**, Manhattan plot anthracycline-related cardiac event (outcome of arrhythmia and heart failure). **C**, *CHST8:KCTD15* Locus Zoom plot anthracycline-related arrhythmia. **C** and **D**, *MMP16* Locus Zoom plot anthracycline-related cardiac event (outcome of arrhythmia and heart failure). *KCTD15* indicates potassium channel tetramerization domain containing 15; and *MMP16*, matrix metalloproteinase 16.

the top associated variant was at the *MMP16* 5' flanking region, rs1094199, reaching genome-wide significance, $P=1.98 \times 10^{-8}$, OR=6.15. At the *CHST8:KCTD15* locus, the sentinel variant (rs12151014) remained the same as in the arrhythmia-only analysis; however, the strength of association was attenuated ($P=2.73 \times 10^{-7}$; OR=2.84) and no longer met the threshold for genome-wide significance (Figure 1B). QQ plots are shown in Figure S2. Given the limited number of events for this composite outcome, these findings should be interpreted as exploratory.

No Overlap of *MMP16*, *CHST8:KCTD15* Arrhythmia-Associated Variants With BMI

We next assessed the *CHST8:KCTD15* and *MMP16* loci for genetic association with other known traits in the NHGRI-EBI Catalog of human genome-wide association studies.^{41,42} Neither locus is previously associated with arrhythmia, heart failure or cardiotoxicity in the GWAS Catalog, but both of these loci have significant associations with BMI.^{43–49} Given the known correlation of arrhythmia and obesity in the literature^{50–52} and that our analyses did not include BMI as a covariate, we examined overlap between variants associated with arrhythmia in this study ($P < 1 \times 10^{-5}$) and variants

associated with BMI in the GWAS Catalog. No overlap was identified between *MMP16* and *CHST8:KCTD15* arrhythmia-associated variants and BMI-associated variants. In sensitivity analyses restricted to participants with available BMI, additional adjustment for BMI did not attenuate the association at the *KCTD15* or the *MMP16* loci; the effect estimate was modestly strengthened (Table S2). We also re-analyzed our arrhythmia GWAS in the MER data set for association with BMI. None of the 24 variants at the *MMP16* locus, which were associated with arrhythmia, $P < 1 \times 10^{-5}$ showed any association with BMI (P values ranging 0.157–0.833). Similarly, at the *CHST8:KCTD15* locus, we did not observe any association ($P < 0.05$) with BMI for variants that were associated with arrhythmia, $P < 1 \times 10^{-5}$, suggesting that the observed associations with doxorubicin-related arrhythmia are independent of BMI.

KCTD15 and *MMP16*, But Not *CHST8* Are Moderately Expressed in Human Heart Tissue and iPSC-Derived Cardiomyocytes

In the GTEx database, *KCTD15* has the highest gene expression in the uterus and skin (median transcripts per million [TPM], 52.0 and 49.64, respectively), but also high expression in artery-tibial, coronary and aorta (TPM,

22.7–37.0). *KCTD15* was expressed at more moderate levels in the heart's left ventricle (median TPM 4.90). The tissue profile for *KCTD15* protein expression was not available in the Protein Atlas, but subcellular localization was described as nucleoplasm and cytoplasm. Expression of *CHST8* in GTEx was predominantly in pituitary tissue (TPM 92.5), followed by brain and breast, with very low expression in any type of heart or arterial tissue (TPM ranging, 0.12–0.89). Across all tissues, *MMP16* was most highly expressed in artery-tibial, coronary and aorta tissues at moderate levels (TPM ranging, 2.6–4.5) with subcellular localization described in the Protein Atlas as cytosolic. We next moved to examine gene expression specifically in available data sets of cardiac cell types.

Gene expression in the heart was assessed for each locus of interest based on RNASeq analysis of human iPSC-derived cardiomyocytes from our previous publication.⁵³ In human iPSC-derived cardiomyocytes, *KCTD15* and *MMP16* were present at moderate levels, with log counts per million of 3.38 and 4.88, respectively, in a data set ranging from log counts per million of –1.43 to 13.83. *CHST8* expression was not observed in this data set.

Given the correlation of arrhythmia with aging,⁵⁴ we also investigated differential gene expression of *CHST8*, *KCTD15* and *MMP16* in the hearts of aged macaque monkeys in a publicly available single cell sequencing data set from the hearts of old and young cynomolgus macaque which annotated 18 different cardiac cell types.⁵⁵ In the hearts of aged macaque monkeys, *MMP16* expression was significantly lower in cardiac endothelial cell types, including arterial, capillary and vascular endothelial cells, with the largest fold change observed in arterial endothelial cells, log fold change = –0.875, adjusted $P = 6.91 \times 10^{-38}$, but not in any other annotated cardiac cell type. Neither *CHST8* nor *KCTD15* were differentially expressed in the aged macaque heart for any cardiac cell type.

KCTD15 and *MMP16* Gene Expression in Response to Doxorubicin in AC16 Cardiomyocytes

We next moved to assess *KCTD15* and *MMP16* gene expression in response to 0.5 $\mu\text{mol/L}$ doxorubicin over 48 hours, using human AC16 cardiomyocytes. In this model, both *KCTD15* and *MMP16* expression were significantly reduced in response to doxorubicin, fold change 2.23, $P < 0.0001$; Figure 2A and fold change 1.44, $P = 0.012$; Figure 2B, respectively.

In Silico Analysis of Associated Variants in the *KCTD15* Promoter Suggested Differential Binding of PAX9 and REST Transcription Factors to the Risk and Wild-Type Haplotypes

Two of the top associated variants at the *CHST8:KCTD15* locus (rs62101381 and rs141116709) mapped 275 bp apart, within the *KCTD15* promoter region (hg38:chr19+:33,795,495–33,797,041) and were in complete LD ($R^2 = 1$). In silico analysis of transcription factor binding affinity to the associated risk and wild-type haplotypes of these variants showed differential binding of the PAX9 transcription factor predicted to have significant binding affinity to the wild-type sequence, $P = 0.004$ and not to the risk sequence, $P = 0.082$, whereas, the NRSF (neuron-restrictive silencer factor; also known as REST) transcription was predicted to have significant binding affinity to the risk sequence, $P = 0.023$ and not significantly to the wild-type sequence, $P = 0.10$ (graphically shown in Figure S3).

MMP16 CRISPR KO in F0 Adult Zebrafish Demonstrated Salutary Modifying Effects to Doxorubicin-Induced Cardiomyopathy

Given the association between *MMP16* and arrhythmia and/or heart failure, we hypothesized that *MMP16* could

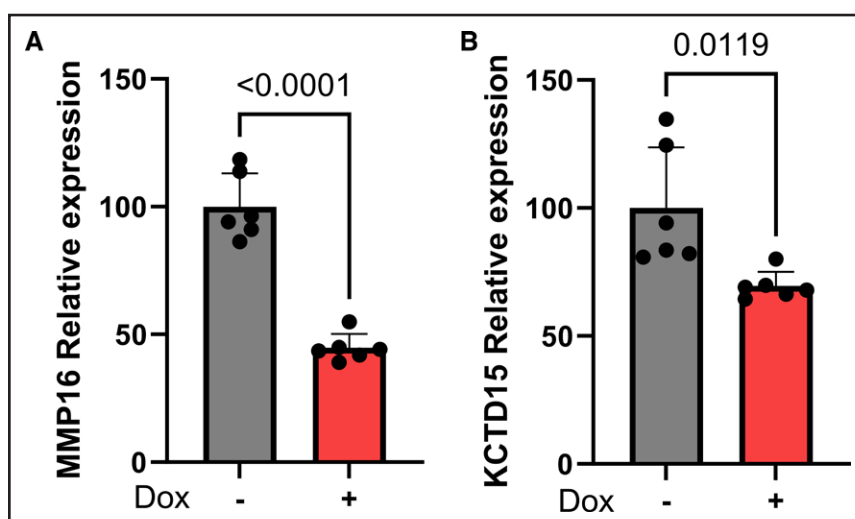


Figure 2. *MMP16* and *KCTD15* gene expression in AC16 cardiomyocytes in response to doxorubicin.

MMP16 (matrix metalloproteinase 16; **A**) and *KCTD15* (potassium channel tetramerization domain containing 15; **B**) gene expression following 48 hours of treatment with 0.5 $\mu\text{mol/L}$ doxorubicin. Data are based on 6 replicates per group. Error bars are SEM.

play a role in AIC. To test this hypothesis, we adopted an adult zebrafish model of AIC for testing the potential modifying effect of a *mmp16* knockout. Due to the duplication of the zebrafish genome, 2 homologous isoforms of *mmp16*, *mmp16a* and *mmp16b*, were identified. We designed sgRNAs for each isoform, which effectively knockout these isoforms at the F0 generation, respectively (Figure 3B and 3C). In contrast to significantly reduced EF in wild type zebrafish treated with doxorubicin ($P=0.003$), EF (EF%) was preserved at levels comparable to those of WT zebrafish without doxorubicin treatment in *mmp16b* (Figure 3A), but not in *mmp16a* mutant fish. These genetic data in zebrafish suggested *mmp16* as a salutary modifier for AIC, and *mmp16b* as the predominant functional isoform of *mmp16* in a zebrafish heart.

DISCUSSION

Defining novel risk genes and variants for anthracycline-induced cardiac outcomes has clinical importance towards the development of personalized medicine approaches. However, barriers are the lack of cohorts with clinical outcome and corresponding

genomic data and the lack of replication due to the winner's curse,⁵⁶ a common problem with the traditional approach of discovery cohort followed by replication in a second (often smaller available sample). We chose a different approach, taking 2 very similar (genetically homogeneous populations) straight to meta-analysis, on the basis that any genome-wide significant hits ($P<5\times 10^{-8}$) would be worthy of further phenotypic exploration and functional validation. We used doxorubicin-related arrhythmia (as it was more prevalent than heart failure) as the primary outcome, and further explored the phenotype by including heart failure in the outcome.

Two novel loci were identified, *CHST8:KCTD15* and *MMP16*, as associated with arrhythmia or anthracycline-related cardiac outcomes. Neither of these has been previously described with this association, likely because we focused primarily on arrhythmia and the majority of genetic association studies of doxorubicin-related cardiotoxicity have used binary outcomes of left ventricular ejection fraction decline by at least 10% points to <50 ,⁵⁷ symptomatic heart failure^{58,59} or quantitative analysis of the maximum decline in left ventricular ejection fraction.^{60,61} To our knowledge, this is the first genetic study of anthracycline-related arrhythmia. Further analyses

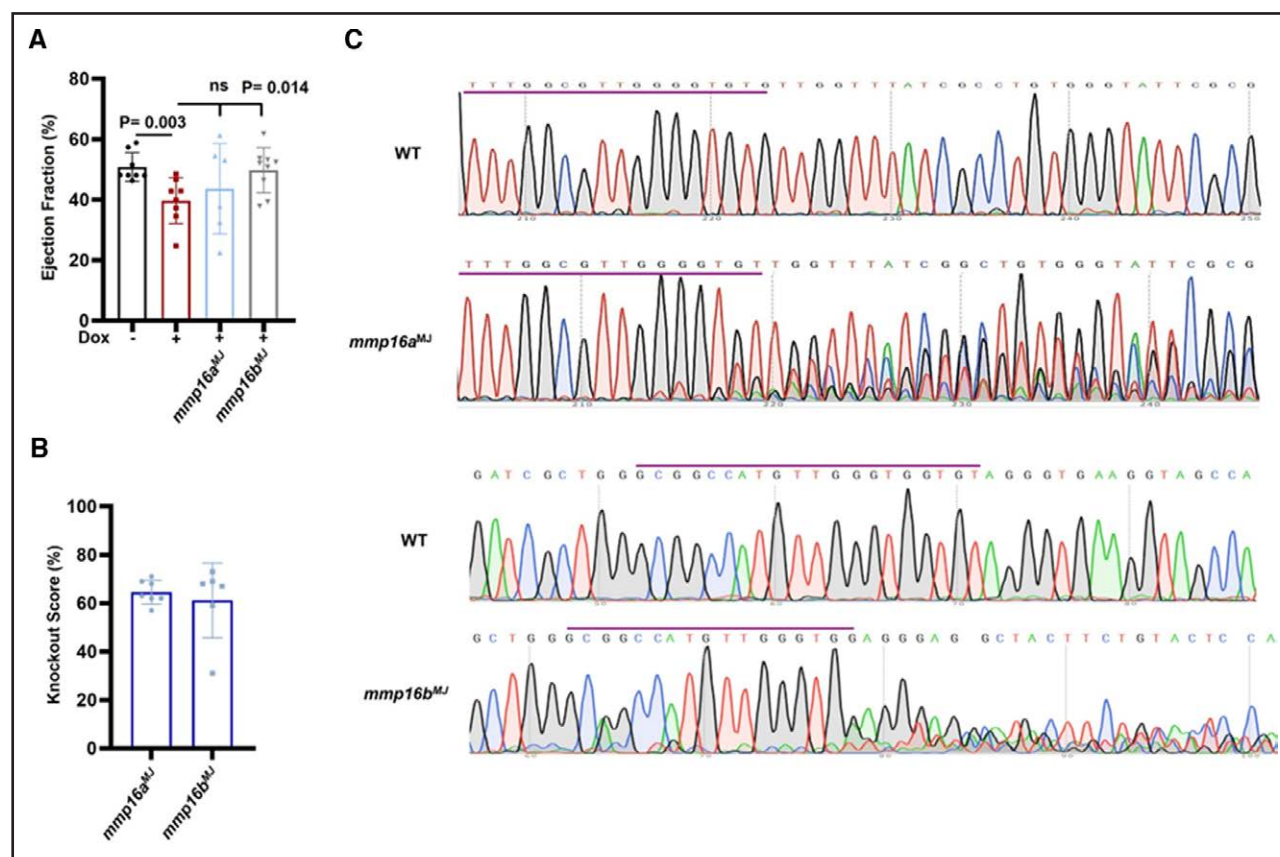


Figure 3. *MMP16b* knockout preserved ejection fraction in an adult zebrafish model of doxorubicin-induced cardiomyopathy.

A, Sanger sequencing wild-type (WT) *mmp16* and knockout of *mmp16a* (*mmp16a*-MJ). **B**, Sanger sequencing WT *mmp16* and knockout of *mmp16b* (*mmp16b*-MJ). **C**, Ejection fraction in wild-type zebrafish $\pm$ doxorubicin and in *mmp16a* and *mmp16b* knockout fish treated with doxorubicin.

of the CHST8:KCTD15 locus focused on *KCTD15*, because *CHST8* is not expressed in cardiomyocytes or other cardiac tissues, and the sentinel variant mapped to the *KCTD15* promoter region with suggestive in silico evidence of the sentinel variant increasing the binding affinity of a transcriptional repressor that is important to cardiac gene expression, discussed below.

KCTD15 belongs to the family of KCTD (potassium channel tetramerization domain) containing proteins, which includes a total of 26 members that share sequence similarity with the cytoplasmic domain of voltage-gated K⁺ channels.⁶² *KCTD15* is ubiquitously expressed in the body, with moderate expression in left ventricular heart tissue (GTEx) and moderate to high expression in our previously published transcriptome data set of human iPSC-derived cardiomyocytes.^{53,63} Little is known about the biological function in normal physiological conditions, but it is known to be overexpressed in acute lymphoid and acute myeloid leukemia⁶⁴ and in HER2+ breast cancer,⁶⁵ and in the SKBR3 HER2+ cell line, *KCTD15* knock-out significantly reduced cell proliferation, cell cycle progression and increased sensitivity to doxorubicin.⁶⁵ In our study, the genetic variants most strongly associated with doxorubicin-related arrhythmia map to the *KCTD15* promoter region, for which in silico analysis suggests could alter the binding affinity of NRSF/REST, a transcriptional repressor, important for regulation of the fetal cardiac gene program.⁶⁶

Mouse models with dominant negative NRSF demonstrate several abnormalities in cardiac electrophysiological properties and ion channel expression^{67,68} and these abnormalities were ameliorated by inhibition of N-type calcium channels, which prevented lethal arrhythmia's in mice with heart failure.⁶⁹ Replication of the *KCTD15* genetic associations and physical validation of the relationship between these variants and NRSF binding are required, but we consider these findings as preliminary data to postulate the hypothesis that genetic variants at the *KCTD15* promoter identify a sub-population of patients at risk for doxorubicin-induced arrhythmia, and inhibition of KCTD15 or N-type calcium channels could be a precision therapy for this group of patients. Given that *KCTD15* silencing in the tumour reduced cell proliferation and increased sensitivity to doxorubicin, inhibition of KCTD15 could also be antineoplastic, allowing lower, less toxic dosing of doxorubicin.

The second locus identified in the primary meta-analysis of doxorubicin-related arrhythmia was *MMP16*, a member of the matrix metalloproteinase family, a group of over 20 genes with biological roles in extracellular matrix remodeling, CVD (particularly atrial remodeling in AF^{70–76}) and cancer.⁷⁷ In mouse models of doxorubicin-induced cardiotoxicity, oral inhibitors of MMP-2 attenuated doxorubicin-induced systolic and diastolic dysfunction⁷⁸ and prevented extracellular matrix remodeling.⁷⁹

Less is known of the relationship of *MMP16* and cardiotoxicity, but there is some evidence of a role for *MMP16* in the biology of AF and cancer, whereby the ratio of *MMP16* and its inhibitor, *TIMP4*, was significantly decreased in AF patients with severe aortic stenosis⁷⁰ and inhibition of *MMP16* via MicroRNA-377-3p inhibited ovarian cancer cell growth, invasion and interstitial transition.⁸⁰ In our study, we observed a significant decrease in *MMP16* expression in human cardiomyocytes (AC16) treated with doxorubicin and in our in vivo model of doxorubicin-induced heart failure, we found that *MMP16* knockout ameliorated the doxorubicin-induced decline in left ventricular ejection fraction. Taken together, these data suggest that a sub-population of patients with *MMP16* genetic variants is susceptible to doxorubicin-related cardiac remodeling, which could potentially be prevented by pharmacological inhibition of *MMP16*, and perhaps that *MMP16* inhibition may have both cardioprotective and anti-neoplastic capabilities.

The main caveats of the study are the relatively small sample size, such that the reported effect sizes of the top hits could be biased by the winner's curse, lack of functional validation at the level of the variant (gene level only) and in vitro and in vivo models did not include assessment of arrhythmia, which was the primary outcome of our genetic screening in humans. In the zebrafish experiments, the 2 *mmp16* orthologs were targeted independently rather than simultaneously, and therefore compensatory upregulation of the remaining orthologue cannot be formally excluded, as has been described following CRISPR/Cas9-mediated mutagenesis in zebrafish.⁸¹ Notably, however, disruption of *mmp16a* alone did not confer a cardiac protective or rescue phenotype, in contrast to *mmp16b*, arguing against simple functional redundancy between the 2 orthologues. Finally, although our current data imply a role of *MMP16* and *KCTD15* in cardiotoxicity, our studies did not elucidate the mechanism, which we consider a future direction. Nonetheless, our study found genome-wide significance of doxorubicin-related arrhythmia in 2 genetically homogeneous patient populations at 2 novel loci, *KCTD15* and *MMP16* and demonstrated in silico and functional validation of their roles in response to doxorubicin, supported by evidence from the literature for these genes as potential therapeutic targets in cancer.

The identification of novel genetic loci associated with anthracycline-related arrhythmia carries significant clinical implications. By detecting CHST8:KCTD15 and *MMP16* as potential risk loci, this study provides a foundation for the development of predictive genetic screening tools that could stratify patients by susceptibility to cardiotoxicity before chemotherapy initiation. Such risk stratification may allow clinicians to tailor cancer therapy regimens, balancing therapeutic efficacy with cardiovascular safety. High-risk individuals could benefit from enhanced cardiac monitoring, early initiation of

cardioprotective therapies, or consideration of alternative, less cardiotoxic agents. As such, understanding the genetic associations of anthracycline-induced arrhythmia may enable mechanistic studies that could inform novel therapeutic targets or preventive strategies. As precision oncology advances, integrating genetic risk factors into cardio-oncology protocols has the potential to significantly reduce treatment-related morbidity and improve long-term outcomes in cancer survivors.

ARTICLE INFORMATION

Received December 15, 2025; accepted June 22, 2026.

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Acknowledgments

Genomic data provided by the Tapestry study were supported by the Mayo Clinic Center for Individualized Medicine. The authors thank all of the participants of the Tapestry study.

Sources of Funding

Research reported in this publication was supported by the National Heart, Lung and Blood Institute of the National Institutes of Health under award numbers 1R01HL169268-03, 1R01HL176555-01A1, and by the Department of Defense under award number W81XWH-22-1-0288/0289 (PR210385). MER (Molecular Epidemiology Resource) is supported by the National Cancer Institute of the National Institutes of Health under Awards P50 CA97274 and U01 CA195568. Project Generation was supported in part by the Center for Individualized Medicine, Mayo Clinic.

Disclosures

None.

Supplemental Material

Supplemental Methods: Genomic Data Generation
Tables S1–S3
Figures S1–S3

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